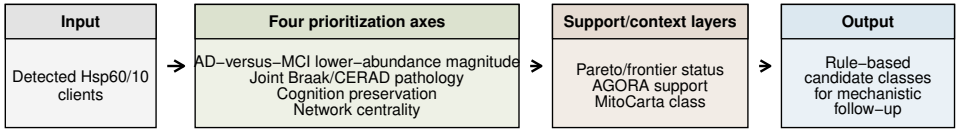
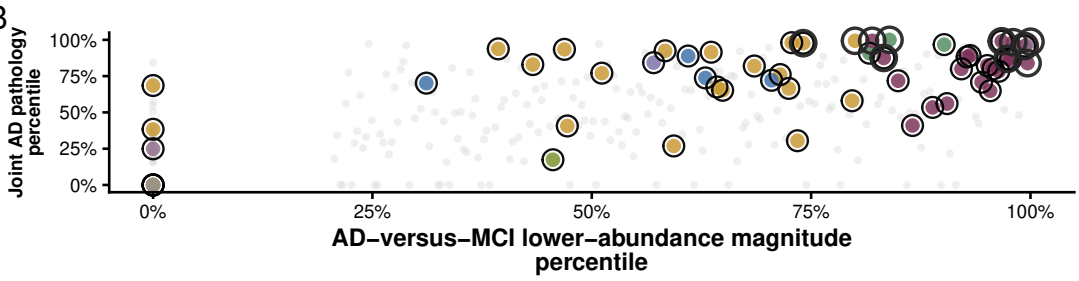
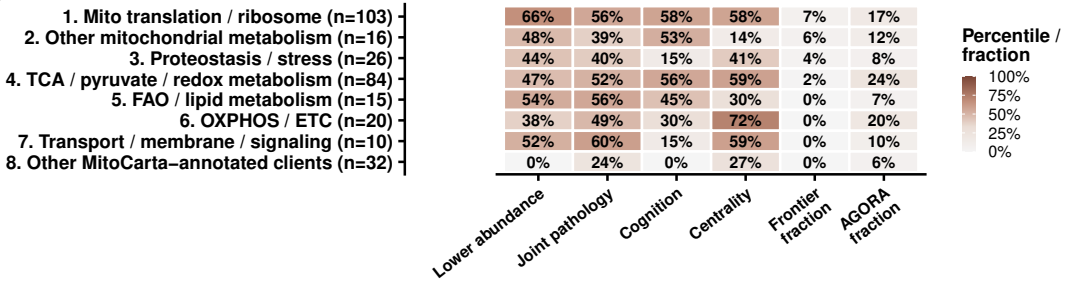
A


Input
Detected Hsp60/10 clients
Four prioritization axes
AD−versus−MCI lower−abundance magnitude
Joint Braak/CERAD pathology
Cognition preservation
Network centrality
Support/context layers
Pareto/frontier status
AGORA support
MitoCarta class
Output
Rule−based candidate classes for mechanistic follow−up

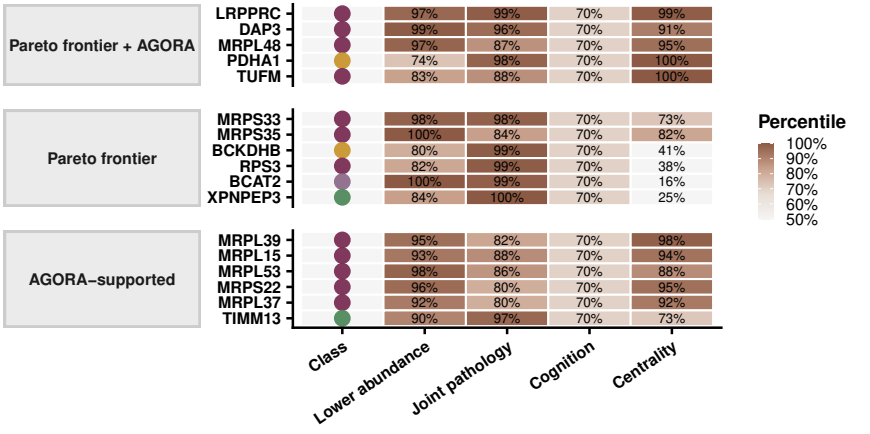
B


Joint AD pathology percentile
100%
75%
50%
25%
0%
0%
25%
50%
75%
100%
AD−versus−MCI lower−abundance magnitude percentile

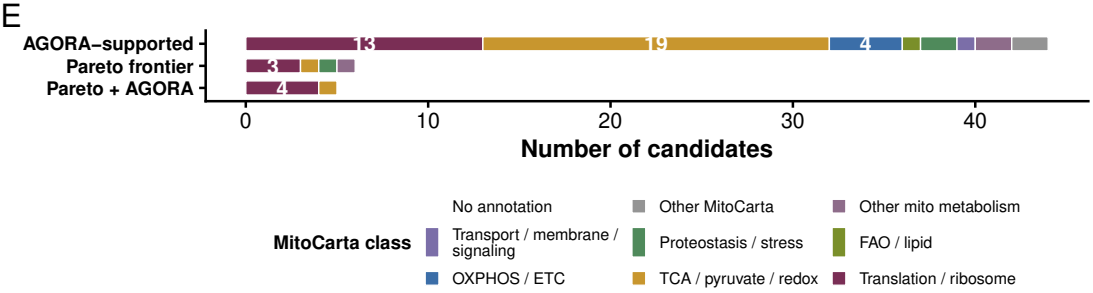
C


1. Mito translation / ribosome (n=103)
2. Other mitochondrial metabolism (n=16)
3. Proteostasis / stress (n=26)
4. TCA / pyruvate / redox metabolism (n=84)
5. FAO / lipid metabolism (n=15)
6. OXPHOS / ETC (n=20)
7. Transport / membrane / signaling (n=10)
8. Other MitoCarta−annotated clients (n=32)
66% 56% 58% 58% 7% 17%
48% 39% 53% 14% 6% 12%
44% 40% 15% 41% 4% 8%
47% 52% 56% 59% 2% 24%
54% 56% 45% 30% 0% 7%
38% 49% 30% 72% 0% 20%
52% 60% 15% 59% 0% 10%
0% 24% 0% 27% 0% 6%
Lower abundance
Joint pathology
Cognition
Centrality
Frontier fraction
AGORA fraction
Percentile / fraction
100%
75%
50%
25%
0%

D


Pareto frontier + AGORA
LRPPRC 97% 99% 70% 99%
DAP3 99% 96% 70% 91%
MRPL48 97% 87% 70% 95%
PDHA1 74% 98% 70% 100%
TUFM 83% 88% 70% 100%
Pareto frontier
MRPS33 98% 98% 70% 73%
MRPS35 100% 84% 70% 82%
BCKDHB 80% 99% 70% 41%
RPS3 82% 99% 70% 38%
BCAT2 100% 99% 70% 16%
XPNPEP3 84% 100% 70% 25%
AGORA−supported
MRPL39 95% 82% 70% 98%
MRPL15 93% 88% 70% 94%
MRPL53 98% 86% 70% 88%
MRPS22 96% 80% 70% 95%
MRPL37 92% 80% 70% 92%
TIMM13 90% 97% 70% 73%
Class
Lower abundance
Joint pathology
Cognition
Centrality
Percentile
100%
90%
80%
70%
60%
50%

E


AGORA−supported
Pareto frontier
Pareto + AGORA
13
19
4
3
4
0
10
20
30
40
Number of candidates
MitoCarta class
No annotation
Other MitoCarta
Other mito metabolism
Transport / membrane / signaling
Proteostasis / stress
FAO / lipid
OXPHOS / ETC
TCA / pyruvate / redox
Translation / ribosome